Lesson plan · for the teacher

How good is my model?

Data science · Level 3 · Lesson 4

Time
40 min

Materials
Board and chalk or markers; Paper and pencils; A phone or tablet with Darsi, for the lab

Learning goals

- See why accuracy can mislead.
- Fill a confusion matrix.
- Choose a threshold.

Activity: A 2 × 2 table

1. Warm-up (5 min): Ask: the forecast says 30% rain. Do you take an umbrella? Does it depend on where you are going?
2. Teach (8 min): Draw the matrix and fill it for the early warning story; show why 90% hid the problem.
3. Count (15 min): Groups fill a matrix from 20 forecast days at a threshold of 0.5, then at 0.3.
4. Decide (7 min): For a school trip and for drying washing, groups choose a threshold and explain.
5. Wrap-up (5 min): Ask two pupils to say the finding in one sentence with a number in it. Then give the exit ticket.

Discussion

- Which mistake is worse for a trip?
- Why did accuracy hide the problem?
- Who should choose the threshold at school?

Exit ticket

- Name the two kinds of mistake and give a school example of each.

Try it at school tomorrow
Draw a 2 × 2 table and fill it with a made-up rain forecast for 20 days. Ask pupils which mistake is worse: an umbrella you didn't need, or getting soaked?

Practice questions · for you

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1 **What happens when we lower the threshold for a warning?**

◯ Nothing changes ◯ More are caught, with more false alarms ◯ Fewer pupils are caught

2 **Match each result to its name.**

Warned, and needed help	• • • •	Miss
Not warned, but needed help	• • • •	False alarm
Warned, but was fine	• • • •	Caught

3 **Count the catches, misses and false alarms, and print them in that order.**

```
needs = [1, 0, 1, 1, 0, 0, 1, 0]
warned = [1, 1, 0, 1, 0, 1, 1, 0]
caught = 0
missed = 0
false_alarm = 0
print(caught, missed, false_alarm)
```

Write your program:

4 **A model that always says "no help" can still have high accuracy.**

◯ True ◯ False

5 **A miss costs a pupil a year; a false alarm costs a short chat. Which threshold suits this?**

◯ A low threshold ◯ No threshold at all ◯ A high threshold

✎ **Think of a decision at school made from a score. Which mistake would be worse there, and who should set the threshold?**

Check your answers

How good is my model?

1 **More are caught, with more false alarms**

A lower bar warns more pupils, rightly and wrongly.

2 **Warned, and needed help — Caught**
Not warned, but needed help — Miss
Warned, but was fine — False alarm

The matrix crosses what the model said with what was true.

3

```
needs = [1, 0, 1, 1, 0, 0, 1, 0]
warned = [1, 1, 0, 1, 0, 1, 1, 0]
caught = 0
missed = 0
false_alarm = 0
for i in range(8):
    if needs[i] == 1 and warned[i] == 1:
        caught = caught + 1
    elif needs[i] == 1:
        missed = missed + 1
    elif warned[i] == 1:
        false_alarm = false_alarm + 1
print(caught, missed, false_alarm)
```

Three pupils were caught, one was missed and two were false alarms.

4 **True**

If few pupils need help, saying no to everyone is right most of the time, yet it misses all of them.

5 **A low threshold**

A low threshold catches more pupils; the extra false alarms are cheap.